

DoangJoo "Alan" Synn

PhD Candidate in Computer Science · Georgia Institute of Technology

Advised by [Prof. HyunJoo Oh](#) and [Prof. Sehoon Ha](#)

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Research Interests

My research builds high-efficiency and scalable machine learning systems, with a focus on automated dataloader tuning, memory-aware and distributed training, and privacy-preserving inference. I work across cloud and distributed computing to make ML training and deployment faster, more resource-efficient, and easier to use.

Education

Georgia Tech, Ph.D. in Computer Science

2022 – Present

- Atlanta, GA, USA
- [School of Interactive Computing](#); advised by Profs. [Oh](#) & [Ha](#)
- Research: computational design, mechanical crafting, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE

2020 – 2022

- Seoul, Korea
- Advised by [Prof. Jong-Kook Kim](#)
- GPA: 4.25 (4.5 scale)
- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

Korea University, B.Eng. in EE

2012 – 2020

- Seoul, Korea
- GPA: 3.56 (4.5 scale)
- Military service leave as a Software Engineer, 2016–2019

Experience

PhD Research Intern, Dolby Laboratories

2024 – 2025

- Atlanta, GA, USA
- Experience Delivery Lab, built a time-series, multimodal datastore supporting low-latency media analytics.

MLOps Engineering Intern, LG Uplus Corp.

2023

- Seoul, Korea
- Led **DeepCrunch**, a model-compression library with multi-backend AI inference for efficient edge serving.

Research Scientist, Protopia AI Inc.

2021 – 2022

- Austin, TX, USA
- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.
- CEO: [Dr. Ebrahimi](#); CTO: [Prof. Esmaeilzadeh](#).

- Seoul, Korea
- Led engineering for 4 startup services, translating ambiguous product needs into back-end systems while mentoring early-career engineers.
- Built production cloud infrastructure with Kubernetes, Terraform, CI/CD, and cost-optimized autoscaling.

Server Engineer, Flysher Inc.

2019

- Seoul, Korea
- Shipped back-end features for 4 social-game titles; one reached global top-10 on Facebook.
- Led an ETL → Redshift analytics pipeline driving data-informed feature rollout.

System Engineer, AKA Intelligence (Musio)

2017 – 2019

- Seoul, Korea
- Designed embedded Linux boards & firmware for the social robot **Musio**.
- Coordinated HW-SW integration and production QA with overseas ODMs.

Selected Publications

- 1 M. Ryu, X. Piao, J. Park, **D. Synn**, and J. Kim, “ADaPT: An Automated Dataloader Parameter Tuning Framework using AVL Tree-based Search Algorithms,” *TKIPS*, Jan. 2025.
- 2 **D. Synn**, X. Piao, J. Park, and J. Kim, “Distributed Training of Deep Learning Models Using Micro Batch Streaming,” *KTSDE*, Mar. 2023.
- 3 X. Piao, **D. Synn**, J. Park, and J. Kim, “Enabling large batch size training for dnn models beyond the memory limit while maintaining performance,” *IEEE Access*, 2023.
- 4 **D. Synn** and J. Kim, “Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders,” *ACK*, Oct. 2021.
- 5 B. Ahn, **D. Synn**, M. Derkani, E. Ebrahimi, and H. Esmailzadeh, “Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection,” *NeurIPS*, 2021.